Momentum analysis: put it together

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use a diagram or calculation to compare a deflection option. Explain where your prediction stops working.

SUCCESS CRITERIA - I can...

1. Compute the required change before evaluating an option.
2. Keep impulse, velocity change and displacement distinct.
3. Give the numerical result and state what the calculation treats as true.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “deflection” and “lead time” to explain your answer. Sentence starters are optional.

deflection: a change in the direction or path of motion

lead time: time available before the modeled event

system: the object or objects chosen for analysis

01 Warm-up: bring back an earlier idea

Total momentum initially is (**15**, **0**) kg·m/s. One fragment leaves with (**3**, **5**) kg·m/s. **Find the other momentum vector** and magnitude.

Word help: magnitude = size of a quantity without its direction

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: shift = Δv × time; J = MΔv.
Fictional calculation that treats the motion as a straight line. M is target mass; J is required impulse. This is not a real encounter forecast.

Worked reasoning: A **1 kg** cart at **+6 m/s** sticks to a **2 kg** cart at rest. Final velocity is **+2 m/s**. The moving cart changes momentum by **-4** and the other by **+4 kg·m/s**.

Your turn: Draw both momentum changes and **show how their sum matches** the system change.

Sentence starter: In my work, I used ... to show ... . I labeled it ... .

Before **After**

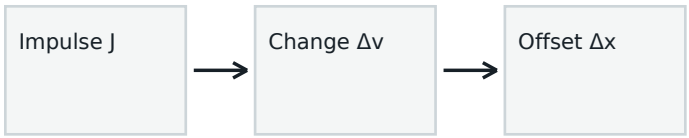
Label mass, velocity and momentum.

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Organize this chain with units: impulse → velocity change → sideways shift. Which step uses target mass?

Sentence starter: I notice ... . This supports ... because ... .



Marzano check - circle: ① ② ③

04 Apply the physics

Target mass **2×10⁹ kg**; desired shift **20 km** over **4×10⁶ s**. **Find required** Δv and impulse. **Compare an option delivering 8×10⁶ N·s**. Organize the known quantities on the before-and-after diagram.

Sentence starter: Both ... . They differ in ... because ... .

Before **After**

Label objects and signed velocities.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

If the target mass estimate is too low, how would a fixed delivered impulse's velocity change compare with the prediction?

Word help: estimate = find an approximate value

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A **4 kg** cart moving right at **6 m/s** stops in **3 s**. Take right as positive. **Find signed impulse** and average force. Use before and after snapshots to organize the momentum account.

Word help: momentum = mass times velocity, **including direction**

Sentence starter: I found … [unit]. This value means … in this situation.

Before	After

Label mass, velocity and momentum.

Marzano check - circle: 1 2 3

07 Exit check

Keep impulse fixed but double both target mass and lead time. **What happens** to predicted shift? Use an equation to **show why**. Organize the known quantities on the before-and-after diagram.

Word help: justify = **support your answer** with a reason or evidence

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart at **+4 m/s** and a **1 kg** cart at **-2 m/s** stick while an external impulse of **-3 N·s** acts. **Find final velocity** and **explain why setting initial momentum** equal to final would fail. Organize the known quantities on the before-and-after diagram.

Sentence starter: For this answer, I treated … as true. I found … . This supports … because … .

Before	After

Label objects and signed velocities.

Marzano check - circle: 1 2 3